Sri Lanka Institute of Information Technology

B. Sc. Honours Degree
in
Information Technology

Final Examination

Year 3, Semester I

IT3020 – Database Systems

Duration: 2 Hours

June 2024

Instructions to Candidates:

- This paper is preceded by a 10-minute reading period. The supervisor will indicate when answering may commence.
- This paper contains 4 questions. Answer All Questions.
- Use the booklets given to provide answers.
- Total marks for the paper will be 100.
- A mark for each question is mentioned in the paper.
- This paper contains six pages with the Cover Pager.
- Electronic devices that store and retrieve text, including calculators and mobile phones are not allowed.

Question 1 **(25 marks)**

Consider the following object relational database schema for recording the assignments of employees to projects:

Object types:
proj_t (pno: char(6), pname: varchar2(12), stdate: date, endate: date)
assignment_t (project: ref proj_t, adate: date, hours: number(2))
assignment_tb: table of assignment_t
emp_t (eno: number(8), ename: varchar2(12), assignments: assignment_tb)

Tables:
projects of proj_t(pno primary key);
emp of emp_t(eno primary key)
nested table assignments store as emp_assigntb;

The attributes of proj_t are project number (*pno*), project name (*pname*), start date (*stdate*), and end date (*endate*). The attributes of assignment_t include ref of proj_t (*project*), date on which the employee was assigned to the project (*adate*), and number of hours per week (*hours*). In the type emp_t, the attributes are employee number (*eno*), name (*ename*), and the set of project assignments of type assignment_tb which is a table of assignment_t. Some sample data for this database is shown below.

PROJECTS

PNO	PNAME	SDATE	EDATE
P01	Lora	01-Jan-2016	01-Jan-2018
P02	Xplore	02-Mar-2015	02-Mar-2016
P03	FLAVer	15-Jun-2017	15-Jun-2020
P04	myGod	20-Apr-2015	20-Apr-2019

EMP

ENO	ENAME	ASSIGNMENTS		
		PROJECT	ADATE	HOURS
EMP0045	Saman Jayawardhana	Lora	01-Jan-16	10
		Xplore	01-Jul-15	15
		myGod	01-Jan-16	12
EMP0078	Kasun Rathnayake	Lora	01-Jan-17	3
		FLAVer	15-Jun-17	8
		myGod	20-Apr-15	20

a) Write Oracle OBJECT SQL statements to answer the following queries (using columns of REF type instead of joins to link tables):

i. For each employee, get the employee number, name, and the total number of hours per week the employee is assigned to work on projects.

(4 marks)

ii. For each project that has employees assigned to it, find the project number, project name, and how many employees are assigned to it.

(5 marks)

b) Assuming that the database contains only the given sample data, assign *Saman Jayawardhana* to project number *P03* for 5 hours per week from 5 November 2017.

(5 marks)

c) Add a member method named *projcnt* to get the number of projects assigned to an employee after a specified date (for example, the number of projects assigned after 1 June 2016). Write Oracle SQL statements to modify the object type emp_t.

(9 marks)

d) Using the method defined above, write an Oracle SQL statement to display the eno, ename, and the number of projects assigned to each employee after 1 January 2016. (A date can be input as a string of the form '01-Jan-16'.)

(2 marks)

Question 2 **(25 marks)**

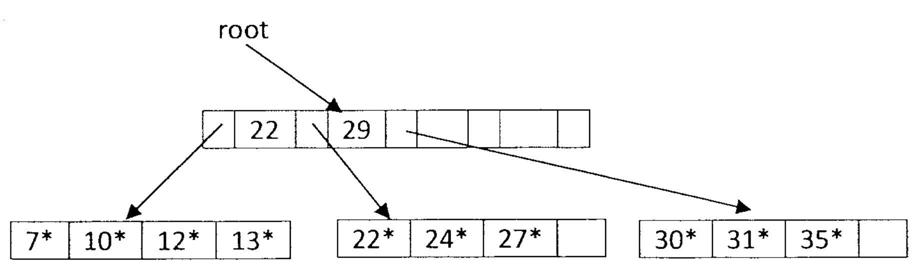
a) Briefly explain three file organization techniques used in database systems to store and manage data efficiently.

(5 marks)

b) Compare Tree based indexing techniques vs hashing based techniques used in database management systems.

(5 marks)

c) Consider the following B+ tree of order 2:



Illustrate the B+ tree after inserting 23 and 26.

(5 marks)

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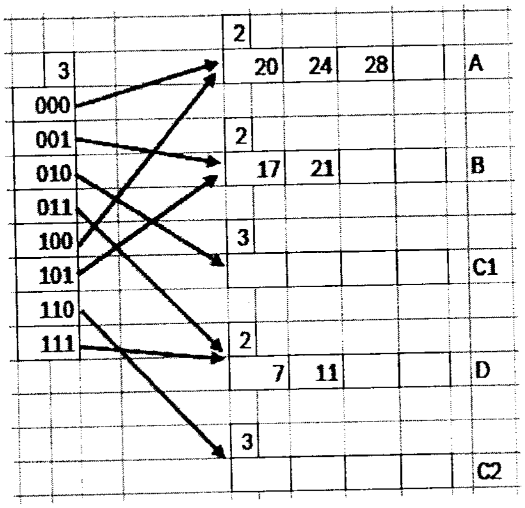
d) Consider the following Extendible Hashed file. Insert 5 suitable sample values into C1 and C2 buckets. The hashing function considers the last digits of the binary representation.



(5 marks)

e) Consider the following relation:

Emp (*eid*: integer, *ename*:string, *age*: integer, *salary*: float)

On this table, there is a dense clustered B+ tree index on *eid* and an unclustered B+ tree index on *age*. The data records are stored in a heap file.

Consider the following query:
SELECT ename, salary
FROM emp
WHERE age > 18;

If 95% of tuples satisfy the selection condition, what would be the best execution plan for processing this query? Justify your answer.

(5 marks)

Question 3 (25 marks)

a) What is the justification for using I/O costs as the main measure to compare different algorithms for evaluating relational operators?

(2 marks)

b) Why does it cost more to use an unclustered index than a clustered index to retrieve tuples that satisfy a range selection?

(3 marks)

c) Briefly describe what happens in external merge sort in first two passes.

(3 marks)

d) Consider the following database schema and answer the questions.

Students (sno, sname, specialization, gpa) 3000 pages with 10 tuple per page.
Units (uno, cname, credits, dept) 400 pages with 10 tuples per page.
Enrolments (uno, sno, grade) 4,000 pages with 100 tuples per page.

i. If there were no indexes on the two tables Units and Enrolments, what would be the cost of joining them using sort merge join algorithm? Assume that there are enough buffer pages to sort each table in 2 passes.

(6 marks)

ii. Consider the following query:

```
SELECT  s.specialization, AVG (s.gpa)
FROM  Students s
GROUP BY  s.specialization;
```

Assume that the Students relation has a clustered B+ tree index on the specialization attribute. Also, assume that the fields in the student relation are of equal size. Additionally, assume that the student records are in sorted order due to the existing clustered index. There are 20 buffer pages available.

What additional index would you create to improve the performance of the above query? Justify your answer. (Hint: Find the cost of plans to justify your answer.)

(11 marks)

Question 4 **(25 marks)**

a) List the properties of a transaction. Briefly explain each of them.

(4 marks)

b) Compare Serial Schedule vs Serializable Schedule?

(2 marks)

c) Briefly explain the rules in Strict 2 Phase Locking (S2PL) Protocol.

(2 marks)

d) Consider a database with objects A and B and assume that there are two transactions T1 and T2. Transaction T1 reads object A and B and then writes object A. Transaction T2 reads objects A and B and then writes objects A and B. Both T1 and T2 commit after all read and write actions of them.

i. Give example schedules with actions of transactions T1 and T2 on objects A and B that results in write-read conflict, read-write conflict and write-write conflict.

(6 marks)

ii. For each of these three schedules, show that Strict 2PL disallows the conflicts.

(2 marks)

e) Consider the following sequence of actions, listed in the order they are submitted to the DBMS. The DBMS processes actions in the order shown. The Strict 2PL is used for concurrency control. If a transaction is blocked, assume that all of its actions are queued until it is resumed; the DBMS continues with the next action of an unblocked transaction.

T1:R(X), T2:W(X), T2:W(Y), T3:W(Y), T1:W(Y),T3W(X), T1: Commit, T2: Commit, T3: Commit

Assume that older transaction has higher priority always.

Draw a transaction schedule and briefly explain how the above three transactions are running till commit those without deadlocks in schedule. Follow *Wait-Die* policy to deal with deadlock.

(5 marks)

f) Briefly explain what the phantom problem is.

(4 marks)

- End of the Question Paper -